

Homework 3

Deadline: September 20, 2026

Problem 1

Using the limit theorems for sequences and the standard limits $\lim_{n \rightarrow \infty} n^{-p} = 0$ for positive integers p , prove that

$$\lim_{n \rightarrow \infty} \frac{11n^6 - 4n^5 + 9n^2 - 7}{13n^6 + 5n^4 + 2n + 1} = \frac{11}{13}.$$

Rewrite the quotient so that every limit theorem you invoke applies.

Problem 2

Suppose that (x_n) and (y_n) are sequences in $\mathbb{R}$ satisfying

$$\lim_{n \rightarrow \infty} x_n = -2, \quad \lim_{n \rightarrow \infty} y_n = 5,$$

and assume that $x_n + 2y_n \neq 0$ for every $n \in \mathbb{N}$. Determine and justify

$$\lim_{n \rightarrow \infty} \frac{2y_n - x_n}{(x_n + 2y_n)^2}.$$

Problem 3

Let (a_n) and (b_n) be sequences in $\mathbb{R}$ such that $a_n \rightarrow a$ and $b_n \rightarrow b$. For each $n \in \mathbb{N}$, define

$$u_n = \frac{a_n^4 - 2a_nb_n + 3}{a_n^2 + b_n^2 + 1}.$$

Prove carefully, using the limit theorems, that

$$\lim_{n \rightarrow \infty} u_n = \frac{a^4 - 2ab + 3}{a^2 + b^2 + 1}.$$

Your proof should explicitly explain why the quotient theorem is applicable.

Problem 4

Let $x_1 = 1$, and define

$$x_{n+1} = 4x_n^2 - x_n \quad (n \geq 1).$$

Prove that $\lim_{n \rightarrow \infty} x_n = +\infty$. You may first establish the estimate $x_n \geq 3^{n-1}$ for every $n \in \mathbb{N}$.

Problem 5

Let (s_n) , (t_n) , and (r_n) be sequences in $\mathbb{R}$. Suppose that $r_n \rightarrow 0$ and that there exists $N_0 \in \mathbb{N}$ such that

$$s_n \leq t_n + r_n \quad \text{for every } n > N_0.$$

Prove directly from the definition that if $s_n \rightarrow +\infty$, then $t_n \rightarrow +\infty$.